The Basin Rule: One Surface-Density Condition for Where Cold Dark Energy Condenses, with Sealed Lensing Predictions and Audited Failures

Working note, 2026-10-28, v1.0. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted; the theory is not closed.

Abstract

This note follows PAPER63 (DOI 10.5281/zenodo.23708507), which asked where cold dark energy switches on. The programme studies a galaxy law $g_N = g^2/(g + a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. It adds a gravitating $w \approx 0$ field component called *cold dark energy*; this is not matter and not a particle.

We report frozen, reproduced calculations. Each was pre-registered and committed before its numbers, run with a mutation twin, and rerun to byte identity. Together they reduce the switch to one condition. A system carries a cored cold dark energy halo iff the collapse surface density of the *gravitational basin* it belongs to beats the galaxy floor:

$$\eta(L)\,\Sigma_{\rm coll}(M_{\rm basin}) > a_0/2\pi G.$$

The same $\max(a_0/2\pi G,\ \eta\Sigma_{\rm coll})$ sets the halo core. The threshold mass is therefore not a separate number: it follows from the concentration level L.

On the canonical footing this rule passes every graded calibration line:

- SPARC;
- 63 groups;
- 15 cluster lensing keys;
- five Local Group dwarf classes;
- KiDS isolated-lens lensing;
- the Milky Way.

Random settings in the explored box pass 10 times in 24.

An X-ray-bright group "wrong sign" is withdrawn as an estimator artefact. A held-out elliptical failure becomes a hot-gas prediction. IC 1613 remains a genuine two-tracer failure.

Two lensing predictions are sealed by content hash. The second is unique to the basin rule: at fixed stellar mass, a central in a massive basin should lens about half as strongly at 0.3 Mpc as an isolated one.

A candidate field equation keys condensation on the expansion of the matter flow. This is not evidence for the framework over ΛCDM.

1 The four equations

The four equations, as graded (canonical footing $a_0 = 9.3603 \times 10^{-11}\,\mathrm{m\,s^{-2}}$; footings never pooled):

(1) $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$.

(2) $g = \nu(g_N/a_0)\,g_N$, with $\nu(y) = \frac{1}{2} + \sqrt{\frac{1}{4} + 1/y}$.

(3) $\Sigma_{\rm core} = \max(a_0/2\pi G,\ \eta(L)\Sigma_{\rm coll})$, with $\eta = \frac{4}{3B(1)}L^{5/4}$ and $B(1) = 3\ln 2 - \pi/2$. A halo exists iff the second term wins on $M_{\rm basin}$. The concentration $c_f(M) = L\,c_{\rm nat}(M)/c_{\rm nat}(10^{14})$ follows the framework's own zoom slope.

(4) The halo holds the native fair share ($\Omega_c/\Omega_b = 5.2715$, from BAO+BBN with no CMB) of the measured baryons, out to the turnaround radius.

A *basin* is the region inside the zero-velocity surface a system belongs to. For the Local Group it is measured from the radial-velocity zero crossing in the Updated Nearby Galaxy Catalog: $R_0 = 794.7 \pm 151.5\,\mathrm{kpc}$, with $\log M_{\rm basin} = 12.392$.

2 Results (each a frozen lane; commit IDs in the repository)

Lane	Frozen verdict and key numbers
BS1 basin switch	Canonical: **resolves the Milky Way contradiction** while keeping every calibration line. The mass switch had needed $T < 10^{11.9}$ for the MW but $T > 10^{12.15}$ for lensing. The CL3 one-number variant ($L = 3.75$, $T = 10^{12.236}$) passes every canonical line.
LEE1 look-elsewhere	**BROAD**: 10/24 random (T, L) draws pass every canonical line (0.417; 68% CP [0.303, 0.539]); alt 0/24. The passing band is $L \approx 3.64$–4.17 with $T \gtrsim 10^{12.19}$. The box was not blind.
FM1 forward model	HG2 bins groups on σ, i.e. on the observed pull. Forward-modelled with the data's own scatter (0.356 dex), the base picture gives $D = -0.492$ against a measured -0.362 ($q = 0.934$): **conditioning explains the sign**. A planted $+0.4$ dex bright-group excess still gives $D < 0$, so the statistic is not diagnostic. The "wrong sign" is **withdrawn**.
SG1 SLUGGS	The IMF lever is at its dynamical bound ($M_\star/M_{\rm JAM} = 0.87$–$0.95$). Measured X-ray gas alone takes the class mean from $+0.152$ to $+0.105$ (M87 closes). Gas extrapolated beyond the measured edges passes at -0.002: **a prediction** for deeper X-ray data.
BC1 X-COP cores	The stellar profiles already carry the BCG. BCG×2 moves the $0.1\,R_{500}$ miss only from $+0.193$ to $+0.168$. The shape failure is the cluster halo's shape, not the baryons.
RT1 / EF1 IC 1613	HI rotation (LITTLE THINGS) gives $E = -0.593$ ($Z = -7.90$), matching the stellar-σ tracer (-0.535). The LG basin's external field shifts it by ≤ 0.0035 dex. **Genuine failure.**
TH1 flow basins	Basins from measured velocity zero crossings reproduce BS1's flags for all 67 Local Volume SPARC galaxies (a weak test). The MW's halo requires MW+M31 baryons $\gtrsim 1.5 \times 10^{11} M_\odot$.

3 Sealed predictions

SP1 (isolated lenses, sha256 `09cf7da6...04bf`). Above $\log M_\star \approx 11.21$, the outer $\Delta\Sigma$ of isolated centrals drops below the law's, with $S_{\rm step} = 0.47$.

SP2 (the basin step, sha256 `75b5fcdb...8b3d`). At fixed $M_\star$ below the own-mass switch, a central in a basin above T carries a cored halo truncated at R_{200}. An isolated central follows the law. The predicted ratio of their own $\Delta\Sigma$ at 0.3 Mpc, averaged over $\log M_\star$ 10.5–11.0, is $S_{\rm env} = 0.484$ (canonical; 0.442 alt). The sign is opposite to halo-model expectations for group centrals.

Both falsifiers are frozen. A KiDS-Legacy or Euclid environment split that isolates the central's own term decides them.

4 A candidate field equation (not derived, not graded)

Every selector keyed on potential depth, σ or acceleration has failed in the record, because satellites sit in shallower potentials than the Sun. A pressure fluid is excluded by the Bullet Cluster.

The basin, as measured, is where the coarse-grained expansion $\theta = \nabla\cdot v$ of the matter flow has turned negative. θ is native to Khronon/AeST-type actions through $c_2(\nabla_\mu u^\mu)^2$. A candidate order parameter χ has

$$m^2 = m_0^2\big[\Theta(\theta_u) - 1 + (a_0/2\pi G)/(\eta\Sigma_{\rm coll})\big],$$

which condenses inside turned-around regions exactly when the $\max(\cdot)$ rule's collapse term wins. An oscillating condensate behaves as cold, collisionless matter. The step function and the smoothing scale are posited. Kill tests are listed in the repository.

5 What is fitted, posited or failing

- **Fitted:** $\kappa = \frac{1}{2}$.
- **Located by data:** $L \approx 3.75$–4. A native test against converged zooms is frozen and pending.
- **Derived from L:** T.
- **Posited:** the basin as the unit, i.e. the field equation above.
- **Failing:**
  - IC 1613;
  - the X-COP cluster shape;
  - the alternative a_0 footing (SPARC and the MW ultra-faints, at every setting).
- **Prediction-dependent:** SLUGGS (outer hot gas).

Nothing here says the data favour the framework over ΛCDM.

AI assistance: Anthropic's Claude (analysis lanes, review, drafting). Data: SPARC; KiDS-1000; eRASS1; X-COP; SLUGGS; UNGC (Karachentsev et al. 2013); LITTLE THINGS (Oh et al. 2015); 2MRS; Kourkchi & Tully 2017. All scripts, frozen criteria and results are in the public repository.